ELEMENTARY LINEAR ALGEBRA

SPRING 2025 EXAM 3 OF 3

"Mathematics is the most beautiful and powerful tool you can use to understand the world."
Roger Penrose

You have worked hard this semester! Apply your mind to all you have learnt these last 8 weeks! Give it your best.!!!

1) **This is a 60 minute Exam. You may use the entire class period if you desire.**
   a) Calculator allowed
   b) No notes allowed
   c) No smart devices/internet allowed

By signing below, I ______________________________ acknowledge that I will uphold academic integrity and only use the resources allowed for each section.

Signature:____________________ Date:____________________

Name:____

Do your scratch work here!

1. a) ***(15pts)*** Determine whether the matrix, $A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$ is diagonalizable. if so diagonalize it.

   b) ***(10pts)*** Construct any matrix D that is **similar** to A. Verify that they are indeed similar (Show that the multiplication PDP^{-1} gives you A).

Do your scratch work here!

2. a) ***(15pts)*** Determine whether $A^{-1} = \begin{bmatrix} \frac{5}{2} & -\frac{1}{2} & -\frac{3}{2} \\ -1 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$ is the inverse of $A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 1 & 3 & 1 \end{bmatrix}$.

b) ***(10pts)*** Determine whether $B^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$ is the inverse of $B = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$.

Do your scratch work here!

3. a) ***(10pts)*** Determine whether $\mathcal{B} = \left\{ \begin{bmatrix} 3 \\ -1 \\ 0 \\ 4 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$ is a basis?

b) ***(15pts)*** From a), if $\mathcal{B}$ is a basis, determine whether it is a basis of $\mathbb{R}^4$. Otherwise, find the subspace for which $\mathcal{B}$ is a basis.

Do your scratch work here!

4. a) ***(15pts)*** Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation defined by:

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 2x + 3y \\ -x + 4y \end{bmatrix}.$$

Find the images under T of $\mathbf{u} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, and $\mathbf{v} + \mathbf{w}$. Sketch the object represented by the four coordinates, and their corresponding image under the transformation.

b) ***(10pts)*** What is the area of of the parallelogram defined by the transformation matrix T in (a)?

Do your scratch work here!

5. Consider the vectors $\mathbf{u} = \begin{bmatrix} \sqrt{3} \\ 1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$.

a) ***(2pts)*** What is the length of $\mathbf{u}$?

b) ***(3pts)*** What is the distance between the vectors $\mathbf{u}$ and $\mathbf{v}$?

c) ***(5pts)*** Are the vectors $\mathbf{u}$ and $\mathbf{v}$ orthogonal? explain your answer.

d) ***(5pts)*** What is the angle between $\mathbf{u}$ and $\mathbf{v}$?

e) Let $\mathcal{B} = \left\{ \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \\ 5 \end{bmatrix} \right\}$, and W be the subspace spanned by $\mathcal{B}$. Let $\mathbf{v} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$.

i) ***(5pts)*** Find the orthogonal projection of $\mathbf{v}$ onto the W.

ii) ***(5pts)*** Write $\mathbf{v}$ as the sum of two orthogonal vectors, one being $\text{proj}_W \mathbf{v}$.

6. a) **(2pts)** A nonhomogeneous equation, $A\mathbf{x} = \mathbf{b}$ is always consistent. True/False.

   b) **(2pts)** A homogeneous equation, $A\mathbf{x} = \mathbf{0}$ is always consistent. True/False.

   c) **(2pts)** A matrix in row-echelon form is unique. True/False.

   d) **(2pts)** If a matrix, A is consistent, then A is invertible. True/False.

   e) **(2pts)** Rotation is a linear transformation. True/False.

   f) **(8pts)** If the equation $A\mathbf{x} = \mathbf{b}$ has a solution, which of the following statements are true?

      (i) $\mathbf{b}$ is in the span of the columns of A.

      (ii) $\mathbf{b}$ is in the span of the rows of A.

      (iii) A is invertible.

      (iv) $\det A \neq 0$.

   g) **(1pt each)** Complete the following:

      (i) $(A^\top)^\top$ =

      (ii) $(AB)^\top$ =

      (iii) $(A + B)^\top$ =

      (iv) $(AB)^{-1}$ =

      (v) $A^{-1}A$ =

      (vi) $(A^{-1})^\top$ =

End of the exam!